

Macroeconomics

Goods and Financial Markets

(Blanchard, Macroeconomics 8th edition Ch. 5-6)

Overview

- ▶ Look at goods and financial markets together, and understand how output and the interest rate are determined in the short run.
- ▶ John Hicks and Alvin Hansen called this the *IS-LM* model.
- ▶ The IS-LM model still represents an essential building block of macroeconomics.
- ▶ Despite its simplicity, the model captures much of what happens in the economy in the short run.

Review

- ▶ We characterized the goods market equilibrium as:

$$Y = Z$$

- ▶ We assumed that consumption was a function of disposable income (income minus taxes), and took investment spending, government spending, and taxes as given:

$$\begin{aligned} Z &= C(Y - T) + \bar{I} + G \\ \Rightarrow Y &= C(Y - T) + \bar{I} + G \end{aligned}$$

- ▶ Using this equilibrium condition, we then looked at the factors that moved equilibrium output, e.g., effects of changes in government spending and of shifts in consumption demand.

Investment

- ▶ Previously, investment was assumed to be constant for simplicity.
- ▶ Investment depends on production Y (or sales) and interest rate i
- ▶ Consider a firm ...
 - ▶ Facing an increase in sales and needing to increase production → need to buy additional machines or build an additional plant.
 - ▶ To buy the new machine, the firm must borrow → higher interest rate makes it less attractive to borrow and buy the machine.
- ▶ Assume that all firms can borrow at the same interest rate, i.e., interest rate on bonds.
- ▶ In reality, many firms borrow from banks, at different rates.
- ▶ Also assume that nominal interest rate = real interest rate.

IS relation

- ▶ Thus, we write investment as:

$$I = I(Y, i)$$

- ▶ And equilibrium in the goods market becomes:

$$Y = C(Y - T) + I(Y, i) + G$$

- ▶ which is the *IS* relation.

Goods Market Equilibrium

- ▶ The demand for goods is an increasing function of output.
- ▶ ZZ is upward-sloping because, for a given value of the interest rate, an increase in output leads to an increase in the demand for goods through its effects on consumption and investment.
- ▶ ZZ is flatter than the 45-degree line: assumed that an increase in output leads to a less than one-for-one increase in demand.
- ▶ Equilibrium requires that the demand for goods be equal to output.
- ▶ Intersection of ZZ and 45-degree line (point A) is the equilibrium level of output.

The Derivation of the IS Curve

Shifts of the IS Curve

Movements Along versus Shifts

- ▶ Movement along the downward-sloping IS curve:
 - ▶ Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output.
- ▶ Shifting the IS curve:
 - ▶ Changes in factors that decrease (increase) the demand for goods given the interest rate shift the IS curve to the left (right).

Financial Markets and the LM Relation

- ▶ Recall the interest rate is determined by the equality of the supply of and the demand for money:

$$M = \$YL(i)$$

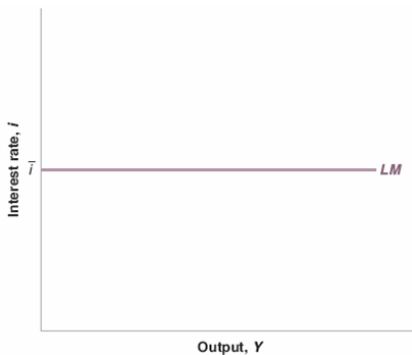
- ▶ Assume central bank sets M .
- ▶ Divide both sides of the equation by the price level P :

$$\frac{M}{P} = YL(i)$$

- ▶ which is the LM relation.
- ▶ In equilibrium, real money supply equals the real money demand, which depends on real income Y , and the interest rate i .

The LM Curve

- ▶ In the past, central banks thought of the money supply as the monetary policy variable.
- ▶ Today, they focus directly on the interest rate, $\bar{i}$, and adjust the money supply so as to achieve it.



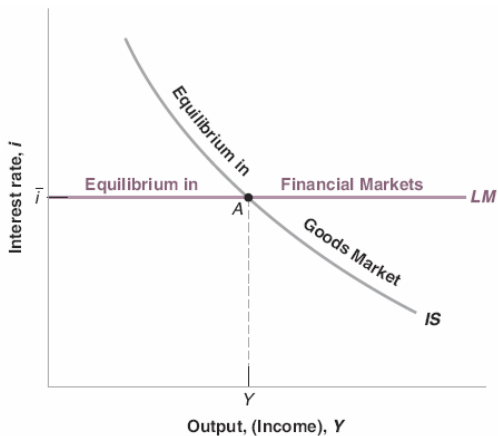
Simultaneous Equilibrium

- ▶ We have

$$\begin{aligned}IS : \quad Y &= C(Y - T) + I(Y, i) + G \\LM : \quad i &= \bar{i}\end{aligned}$$

- ▶ The IS and LM relations together determine output.
- ▶ Only at point A, which is on both curves, are both goods and financial markets in equilibrium.

Simultaneous Equilibrium (cont.)

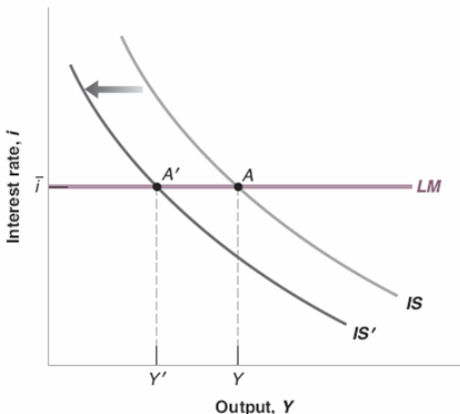


Fiscal policy

- ▶ Decrease in $G - T \Leftrightarrow$ fiscal contraction $\Leftrightarrow$ fiscal consolidation
- ▶ Increase in $G - T \Leftrightarrow$ fiscal expansion
- ▶ Steps to analyze effects of changes in policy, exogenous variables:
 - ▶ Does it shift the IS curve and/or the LM curve?
 - ▶ What does this do to equilibrium output and the equilibrium interest rate?
 - ▶ Describe the effects in words.

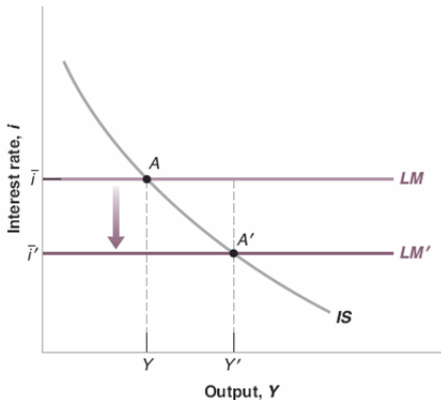
The Effects of an Increase in Taxes

- ▶ An increase in taxes shifts the IS curve to the left.
- ▶ This leads to a decrease in the equilibrium level of output.



Monetary policy

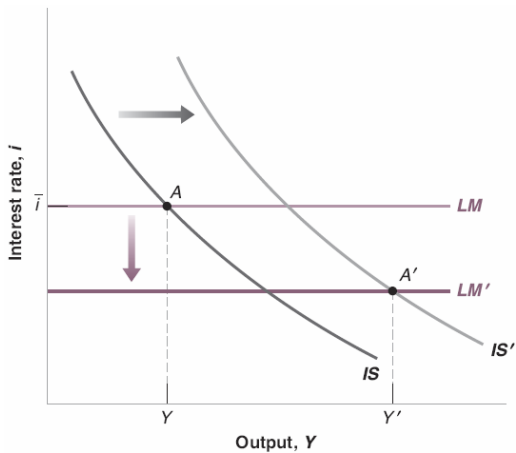
- ▶ Decrease in $i \Leftrightarrow$ increase in $M \Leftrightarrow$ monetary expansion
- ▶ Increase in $i \Leftrightarrow$ decrease in $M \Leftrightarrow$ monetary contraction/tightening
- ▶ A monetary expansion shifts the LM curve down, and leads to higher output.



Using a Policy Mix

- ▶ Monetary-fiscal policy mix is the combination of monetary and fiscal policies.
- ▶ Suppose that the economy is in a recession and output is too low.
- ▶ Both fiscal and monetary policies can be used to increase output.
- ▶ The fiscal expansion shifts the IS curve to the right.
- ▶ A monetary expansion shifts the LM curve down.
- ▶ Both lead to higher output.

Using a Policy Mix (cont.)



Why use a mix?

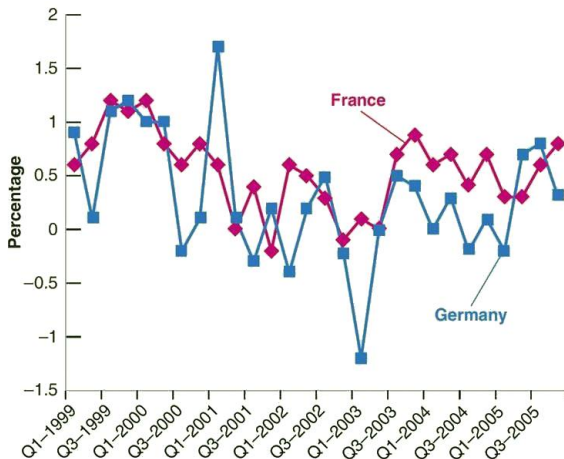
- ▶ Fiscal and monetary policies have different effects on the composition of output.
- ▶ e.g., decrease in income taxes tends to increase consumption relative to investment, decrease in interest rate affects investment more than consumption.
- ▶ Depending on the initial composition of output, policy makers may want to rely more on fiscal or more on monetary policy.
- ▶ Neither fiscal policy nor monetary policy work perfectly. A decrease in taxes may fail to increase consumption. A decrease in the interest rate may fail to increase investment.
- ▶ In case one policy does not work as well as hoped for, it is better to use both.

German and French Recessions of 2001-02

- ▶ Around 2000, several industrial nations suffered from a decline in economic activity; factors:
 1. After prolonged economic boom + low inflation (1990s), inevitable that a slump in economic activity and employment would follow.
 2. Asian financial crisis and the banking crises that hit many emerging market economies took their toll on international trade, hence reducing exports of industrial nations.
 3. Collapse of the dot-com or information technology bubble in 2001 further impacted the economies of developed nations negatively.
- ▶ Both manufacturing investments and consumption expenditures of private households declined, weakening GDP growth in 2001-02.
- ▶ The only sector that showed signs of stabilization in Europe was the construction sector.

Quarterly GDP Growth Rates (1999-2005)

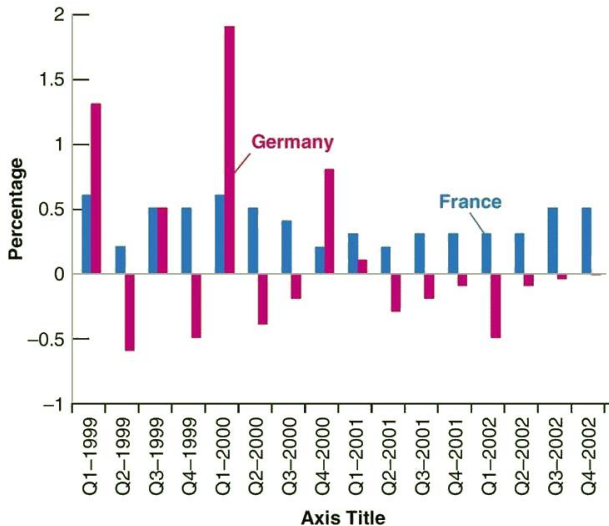
- ▶ Neither Germany nor France witnessed two consecutive quarters of negative GDP growth.



Launch of euro

- ▶ As it launched the euro in January 1999, the EU was in a state of cautious anticipation during the early 2000s after the recessions in both of these nations had led to a decline in the value of the euro.
- ▶ Not till 2002 that the value of the euro began to increase, reaching parity with the US Dollar in mid-2002.
- ▶ Stronger euro hurt European exports.
 - ▶ French exports declined by 2.4% 2001 Q4.
 - ▶ German exports declined for the first time in decades by -0.1% in 2002 Q1.
- ▶ By mid-2002 both nations claimed that their recessions had ended. In reality, however, the economic slump persisted until mid-2005.
- ▶ Slowdown was partly blamed on weak macroeconomic policies.

Government Expenditure (1999-2002)



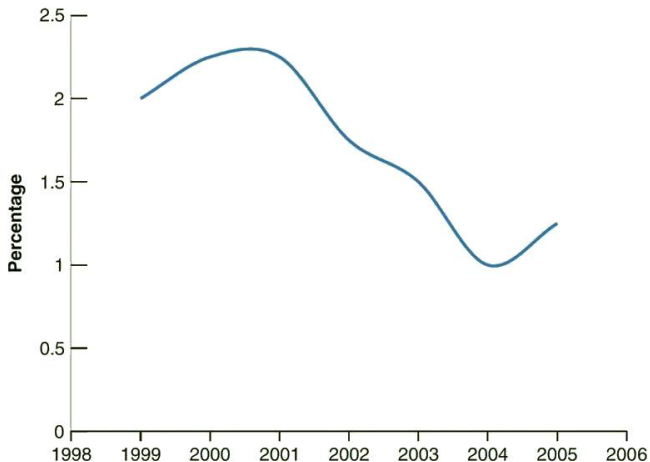
Fiscal Policy

- ▶ Response of the cut in fiscal spending was more marked in Germany in comparison to France.
- ▶ France committed to high social welfare spending (welfare state).
- ▶ On the other hand, Germany's Social-Democrats/Greens ruling coalition at the time introduced the socially unpopular fiscal austerity and labor reform package.
- ▶ The German Chancellor implemented fiscal measures aiming to reduce the burdens and pressures on the fiscal budget exerted by the generous welfare system.
- ▶ Reforms phased out from 2003-10: gradual tax increases, phased rises in pension contributions, cuts in pension benefits, reductions in the amounts allocated to medical treatment, and cutbacks in unemployment benefits.

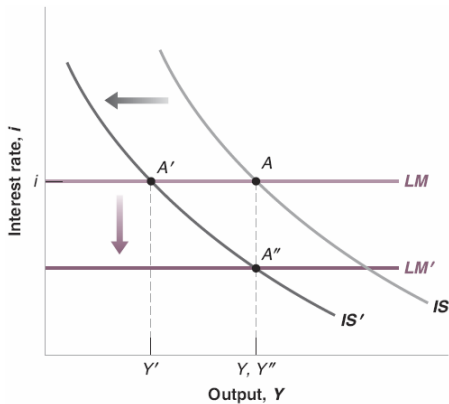
ECB Response

- ▶ Economic conditions of France and Germany are important because they constitute more than 50% of EU's GDP.
- ▶ Sensing the potential impact of the recession on the rest of the Euro zone, the European Central Bank (ECB) adopted an expansionary monetary policy:
 - ▶ Increased money supply and decreased the interbank rate aggressively.
 - ▶ Successful in effectively boosting consumption and investment expenditures.
- ▶ Monetary and fiscal policy mix was used to reduce the impact of the recession in both France and Germany.

ECB Euro Interbank Rates (1999-2005)



Fiscal Consolidation and Monetary Expansion



Deficit Reduction: Good or Bad for Investment?

- ▶ True or False?
- ▶ Private saving goes either toward financing the budget deficit or financing investment. Therefore, reducing the budget deficit leaves more saving available for investment, so investment increases.

Deficit Reduction: Good or Bad for Investment?

- ▶ Answer: False!
- ▶ Recall that equilibrium in the goods market implies:

$$\begin{array}{rccccccc} \text{Investment} & = & \text{Private saving} & + & \text{Public saving} \\ I & = & S & + & (T - G) \end{array}$$

- ▶ Given private saving (S), a lower government deficit (higher $T - G$) means higher I .
- ▶ However, a fiscal contraction lowers output and so S goes down by more than $T - G$ increases, so I decreases.

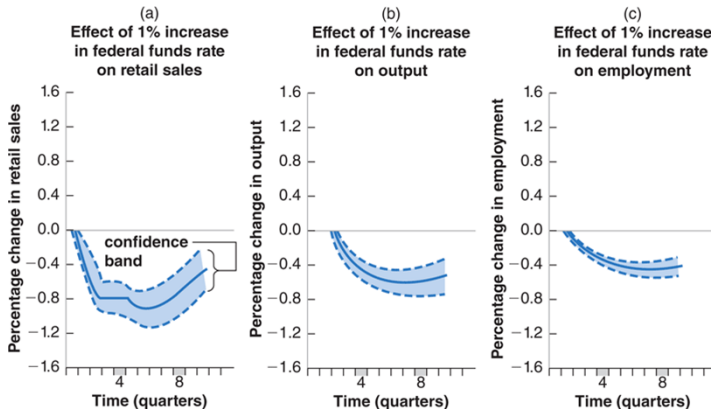
Deficit Reduction: Good or Bad for Investment?

- ▶ Does this mean that deficit reduction always decreases investment?
- ▶ Not necessarily: If when the deficit is reduced, the central bank also decreases the interest rate so as to keep output constant, then investment necessarily goes up.
- ▶ Although output is unchanged, the lower interest rate leads to higher investment.
- ▶ Conclusion: Whether deficit reduction leads to an increase in investment is far from automatic. It may or it may not, depending on the response of monetary policy.

How Does the IS-LM Model Fit the Facts?

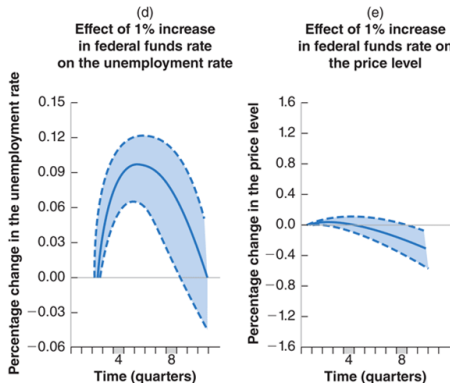
- ▶ Adjustment of output takes time: reintroduce dynamics
 - ▶ Consumers are likely to take time to adjust their consumption following a change in disposable income.
 - ▶ Firms: investment from change in sales, interest rate
 - ▶ Firms: production from change in sales.
- ▶ Time is needed for output to adjust to changes in fiscal and monetary policy. How much time?
- ▶ This question can only be answered by looking at the data and using econometrics.

Empirical Effects of Increase in Federal Funds Rate



Source: Lawrence Christiano, Martin Eichenbaum, and Charles Evans, "The Effects of Monetary Policy Shocks: Evidence From the Flow of Funds," *Review of Economics and Statistics*. 1996, 78 (February): pp. 16–34.

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Empirical Effects of Increase in Federal Funds Rate

- ▶ Largest decrease in retail sales, -0.9% , is achieved after 5 quarters.
- ▶ In response to the decrease in sales, firms cut production, but by less than the decrease in sales, i.e., firms accumulate inventories for some time.
- ▶ The adjustment of production is smoother and slower than the adjustment of sales.
- ▶ The largest decrease, -0.7% , is reached after 8 quarters.
- ▶ In other words, monetary policy works, but it works with long lags: nearly two years to have its full effect on output.

Empirical Effects of Increase in Federal Funds Rate

- ▶ Decline in employment is slow and steady, reaching -0.5% after 8 quarters.
- ▶ Assumption of fixed prices is not a bad approximation of reality in the short run:
 - ▶ Price level is nearly unchanged for the first six quarters or so.
- ▶ This gives us a strong hint as to why the IS-LM model becomes less reliable as we look at the medium run:
 - ▶ In the medium run, we can no longer assume that the price level is given, and movements in the price level become important.

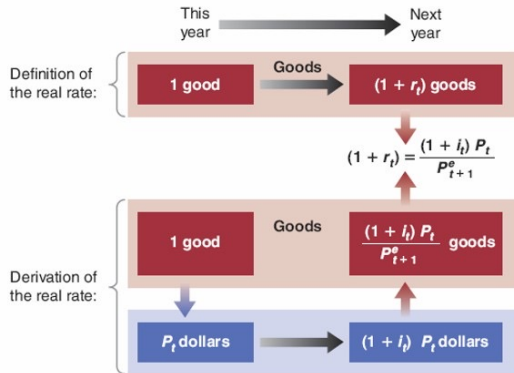
IS-LM and Financial Markets

- ▶ Until now, we assumed that there were only two financial assets (money and bonds), and just one interest rate (the rate on bonds) determined by monetary policy.
- ▶ Financial system also plays a major role in the economy (e.g., 7% of US GDP)
- ▶ Before the 2008 crisis, the importance of financial system was downplayed in macroeconomics.
 - ▶ All interest rates were often assumed to move together with the rate determined by monetary policy.
- ▶ The crisis made painfully clear that this assumption was too simplistic and that the financial system can be subject to crises with major macroeconomic implications.

Interest Rates

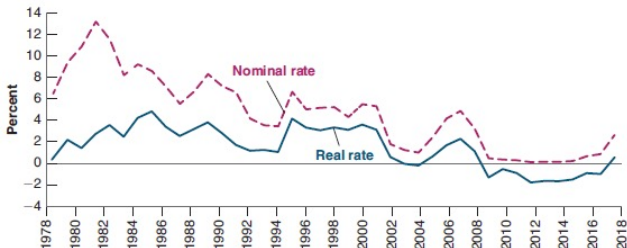
- ▶ Nominal interest rate is the interest rate in terms of dollars.
- ▶ Real interest rate is the interest rate in terms of a basket of goods.
- ▶ We must adjust the nominal interest rate to take into account expected inflation.
- ▶ e.g., 1980: one-year US T-bill rate 10.9%, expected inflation 9.5%
- ▶ 2006: one-year US T-bill rate 4.2%, expected inflation 2.5%

Nominal versus Real Interest Rates



Data: Nominal and Real One-Year T-Bill Rates

- ▶ The nominal interest rate has declined considerably since the early 1980s, but because expected inflation has declined as well, the real rate has declined much less than the nominal rate.

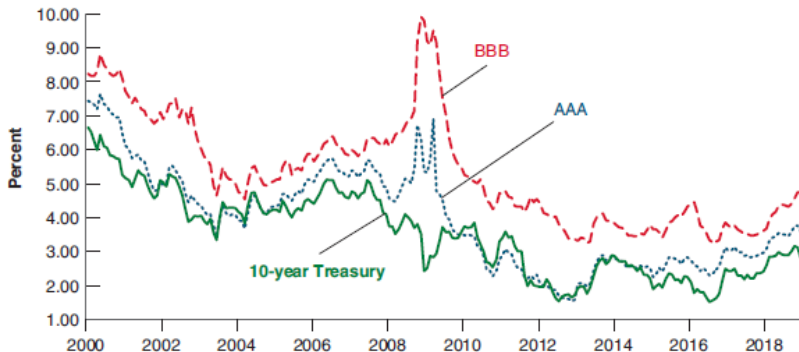


Risk premium

- ▶ Some bonds are risky, so bond holders require a risk premium.
 - ▶ e.g., ordinary people cannot borrow at federal funds rate set by Fed
- ▶ Risk premia are determined by:
 - ▶ The probability of default
 - ▶ The degree of risk aversion of bond holders
- ▶ Let i be the nominal interest rate on a riskless bond, x be the risk premium, and p is the probability of defaulting.
- ▶ To get same expected return on risky bonds as riskless bond:

10-Year Yields

- ▶ In September 2008, the financial crisis led to a sharp increase in the rates at which firms could borrow.



The Role of Financial Intermediaries

- ▶ Until now, we have looked at direct finance: borrowing directly by the ultimate borrowers from the ultimate lenders.
- ▶ In fact, much of the borrowing and lending takes place through financial intermediaries: financial institutions that receive funds from investors and then lend these funds to others.
- ▶ e.g., bank, and non-banks like mortgage companies, money market funds, hedge funds
- ▶ They perform an important function: develop expertise about specific borrowers and can tailor lending to their specific needs.
- ▶ They borrow and lend, charging a slightly higher interest rate than the rate at which they borrow so as to make a profit.

Bank Assets, Capital, and Liabilities

- ▶ Assets: reserves (central bank money), loans to consumers, loans to firms, loans to other banks, mortgages, government bonds, or other forms of securities.
- ▶ Liabilities: checkable deposits, interest-paying deposits, or borrowing from investors and other banks.

Bank Balance Sheet

Assets 100	Liabilities 80 Capital 20
------------	------------------------------

- ▶ *Capital ratio* (capital to assets) = $20/100 = 20\%$
- ▶ *Leverage ratio* (assets to capital) = $100/20 = 5$

Example

- ▶ Expected rate of return on assets = 5%; liabilities = 4%

Leverage and Solvency

- ▶ A higher leverage ratio implies a higher expected profit rate, but also implies a higher risk of insolvency and bankruptcy.
 - ▶ A bank is *solvent* if the value of its assets exceeds the value of its liabilities; otherwise, *insolvent*.
- ▶ Higher leverage also implies a higher risk that the value of the assets becomes less than the value of its liabilities, which, in turn, implies a higher risk of insolvency.
- ▶ For bank with liabilities = 80, its assets can decrease in value down to 80 without the bank becoming insolvent and going bankrupt.
- ▶ But if it were to choose a leverage ratio of 10, any decrease in value of assets below 90 would lead the bank to become insolvent.
 - ▶ Leverage too low → less profit.
 - ▶ Leverage too high → risk of bankruptcy.

Leverage and Solvency (con.)

Liquidity

- ▶ If investors have doubts about the value of the bank assets, the safe thing for them to do is to take their funds out of the bank.
- ▶ This creates serious problems for the bank, which needs to find the funds to repay the investors.
- ▶ The loans it has made cannot easily be called back, e.g., firm has purchased machinery → maturity mismatch!
- ▶ Selling the loans to another bank is likely to be difficult as well.

Liquidity (cont.)

- ▶ The lower the liquidity of bank assets means the more difficult they are to sell:
 - ▶ The higher the risk of being sold at **fire sale prices** (prices far below the true value) and the risk that the bank becomes insolvent.
- ▶ The higher the liquidity of the liabilities (e.g., checkable or demand deposits):
 - ▶ The higher the risk of fire sales, and the risk that the bank becomes insolvent and thus faces **bank runs**.
 - ▶ Danish deposit insurance: “The Guarantee Fund covers registered deposits up to an amount equivalent to EUR 100,000 per depositor.”

Bank Runs

- ▶ The U.S. financial history up to the 1930s is full of bank runs.
- ▶ One potential solution to bank runs is narrow banking, which restricts banks from making loans, and to hold liquid and safe government bonds.
- ▶ To limit bank runs, the United States introduced federal deposit insurance in 1934 (FDIC), \$250,000 in 2008.
- ▶ <https://www.youtube.com/watch?v=sKjdT8I6TnE>
- ▶ https://www.youtube.com/watch?v=QACGoKb48_0
- ▶ <https://www.bbc.com/news/business-65370751>

Extending the IS-LM model

- ▶ Extend the IS-LM model to reflect the distinction between:
 - ▶ Nominal interest rate and real interest rate
 - ▶ Policy rate set by central bank and interest rates faced by borrowers

Financial Shocks, Monetary Policy, and Output

U.S. Housing Prices since 2000

- ▶ The increase in housing prices from 2000 to 2006 was followed by a sharp decline thereafter.
- ▶ Most economists forecast that this would lead to a decrease in demand and a slowdown in growth.
- ▶ Few economists anticipated a major macroeconomic crisis.
- ▶ Case-Shiller Home Price Index:



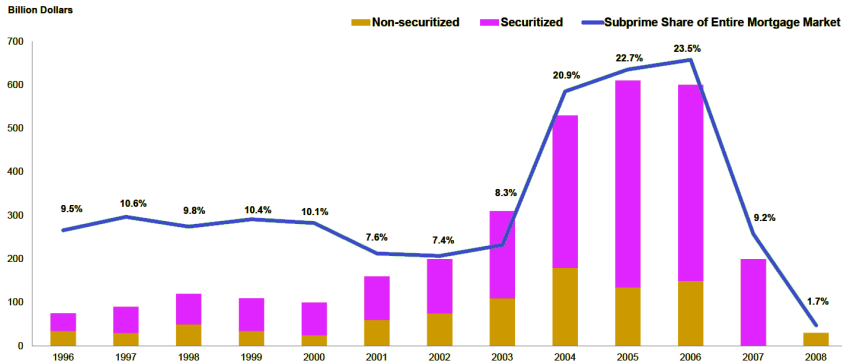
Mortgages

- ▶ The 2000s were a period of unusually low interest rates, which stimulated housing demand.
- ▶ Mortgage lenders were increasingly willing to make loans to risky borrowers with subprime mortgages, or subprimes.
- ▶ Under assumption of rising prices, value of mortgage would decrease over time relative to house price.
- ▶ From 2006 on, many home mortgages went underwater (when the value of the mortgage exceeded the value of the house).
- ▶ Lenders faced large losses as many borrowers defaulted.

Subprime Mortgage

- ▶ Example: borrower has poor credit history, no steady work history
- ▶ Challenge for lender: loan will not conform, need to take the risk; set high interest rate
- ▶ But, can refinance loan, and contractually incur a penalty: (pre-payment) penalty effectively transfers some of home appreciation back to the lender
- ▶ Subprime mortgage only makes sense if everyone expects housing prices to rise
- ▶ If so, can be useful financial innovation: allows borrowers to share potential upside of their housing investment to get a lower overall rate

Subprime Mortgage Originations



Note: Percent securitized is defined as subprime securities issued divided by originations in a given year. In 2007, securities issued exceeded originations.
Source: Inside Mortgage Finance.

From a Housing Problem to a Financial Crisis

- ▶ In mid-2008, losses on mortgages were estimated to be ~\$300 billion: not large *relative* to size of US economy (2% of GDP).
- ▶ Financial system should be able to absorb the shock with limited adverse effect on output.
- ▶ Although trigger of crisis was indeed the decline in housing prices, effects were enormously amplified by financial intermediaries.
- ▶ Banks were highly leveraged because:
 - ▶ Banks probably underestimated the risk,
 - ▶ Bank managers had incentives to go for high expected returns without fully taking the risk of bankruptcy,
 - ▶ Banks avoided financial regulations with structured investment vehicles (SIVs).

SIVs

- ▶ Liability side: borrowed from investors, typically short-term debt
- ▶ Asset side: various forms of securities, typically also guarantee from bank to provide funds to SIV.
- ▶ Main reason for separate vehicle: increase leverage! Operations on balance sheet subject to regulatory capital requirements.
- ▶ Operations through SIV did not require banks to put down capital.
- ▶ When housing prices started declining and many mortgages turned out to be bad, the securities held by SIVs dropped in value.
- ▶ Questions arose about the solvency of the SIVs, and given the guarantee by banks to provide funds to the SIVs if needed, about the solvency of the banks themselves.

Securitization

- ▶ Securitization: creation of securities based on a bundle of assets, such as mortgage-based securities (MBS).
- ▶ Senior securities have first claims on the return from the bundle of assets; junior securities come after.
- ▶ Securitization was a way of diversifying risk, but it came with costs:
 - ▶ The bank that sold the mortgage had few incentives to keep the risk low.
 - ▶ Even for toxic assets, the risk is difficult for rating agencies to assess. It led investors to assume the worst and be reluctant either to hold them or to continue lending to those institutions that did hold them, i.e., illiquid!

U.S. Consumer and Business Confidence, 2007-2011

- ▶ Borrowing became very expensive.
- ▶ Sharp drop in confidence, which bottomed in early 2009.



Policy Responses

- ▶ Financial policies:
 - ▶ Federal deposit insurance was raised from \$100,000 to \$250,000,
 - ▶ The Fed provided widespread liquidity to the financial system through liquidity facilities, and increased the number the assets that could serve as collateral,
 - ▶ The government introduced the Troubled Asset Relief Program (TARP): purchase of shares in banks (i.e., increase capital).

Policy Responses (con.)

- ▶ Monetary policy:

- ▶ The federal funds rate was down to zero by December 2008.
- ▶ The Fed also used unconventional monetary policy, which involved buying other assets as to directly affect the rate faced by borrowers.

- ▶ Fiscal policy:

- ▶ The American Recovery and Reinvestment Act was passed in February 2009, calling for \$780 billion in tax reductions and spending increases

Policy Responses (con.)